

Hampus Serneke

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EDUCATION

École polytechnique fédérale de Lausanne, EPFL

M.Sc. in Data Science

Lausanne, Switzerland

Sept. 2026 – July 2028

- **Ongoing coursework:** machine learning, mathematics of data, measure theory, foundations of data science, ethics and law of AI, French

Lund University

B.Sc. in Technology as part of the Engineering Mathematics program

Lund, Sweden

Aug. 2022 – June 2026

- **Cumulative GPA:** 4.9/5.0
- Pursued a major in Computer Science and Engineering for two years before transferring to Engineering Mathematics.
- **Thesis title:** Deep Learning for Rotation Averaging in Structure from Motion (GNNs in computer vision)
- **Coursework:** Mathematical statistics, matrix theory, stationary stochastic processes, applied mathematics (PDEs), algorithms and data structures, functional programming, object-oriented modelling, electromagnetic field theory

Lund University

Theoretical Philosophy, 30 ECTS

Lund, Sweden

Aug. 2021 – Jan. 2022

- **GPA:** VG/VG
- **Coursework:** Philosophy of science, philosophy of mind, philosophical method, history of philosophy
- Studied philosophy during a gap year while working part-time as a substitute teacher

Katedralskolan Lund

Natural Science High School program

Lund, Sweden

Aug. 2018 – June 2021

- **GPA:** 22.5/22.5

WORK & LEADERSHIP EXPERIENCE

California Institute of Technology, Caltech

Summer Undergraduate Research Fellow

Pasadena, USA

June 2026 – Aug. 2026

- Conducted research in reinforcement learning and AI safety at the Advanced Mechanical Bipedal Experimental Robotics Lab under the supervision of Dr. Pol Mestres and Prof. Aaron Ames
- Created the first hardware implementation of an anytime safe reinforcement learning algorithm (RL-SGF), going from theorem to hardware
- Ran hardware experiments on G1 humanoids and Crazyflie quadrotors using policies trained in IsaacLab and mjlab
- Performed system identification for Crazyflie quadrotors
- Implemented a novel learning rate scheduler for fulfilling the safety requirements, increasing mean reward by 45%
- Obtained a sharper theoretical bound for the required learning rate to ensure safety
- Investigated the effects of relaxing the theory by introducing PPO-inspired estimates

Lund University

Teaching Assistant in Mathematics, Faculty of Engineering

Lund, Sweden

Aug. 2025 – Mar. 2026

- Led weekly exercise sessions where I taught calculus to a class of 30 engineering undergraduates

Axis Communications

Machine Learning Intern, Core Technologies Analytics

Lund, Sweden

June 2023 – May 2025

- Reduced bias in computer vision models using synthetic data
- Developed a pipeline for generating synthetic data in Unreal Engine and independently designed experiments for testing its effectiveness
- Trained deep learning models for object detection and improved overall mAP, while reducing coverage bias by greatly improving performance on key object classes
- Implemented a novel noise reduction algorithm for pre-processing synthetic data in Python
- Streamlined the rendering pipeline in Unreal and refactored the codebase in C++
- Developed a Python package for functional handling of training result files
- Created a visualization tool in Tkinter
- Collaborated across departments to generate data for different applications, onboarded consultants, and worked directly with a team of 10 engineers
- Worked part-time during the terms and full-time during the summers

Hemkodat

Coding Instructor

Sweden

Sept. 2018 – Jan. 2023

- Travelled in Sweden during holidays and weekends to deliver courses in Python, Lua, and C# for adolescents
- Contributed to onboarding and training of new instructors

Internationella Engelska Skolan*Substitute Teacher***Lund, Sweden**

Sept. 2018 – Jan. 2022

- Worked part-time as a substitute teacher and taught in English.

The Computer Engineering Guild at LTH*Vice President***Lund, Sweden**

Jan. 2023 – Dec. 2023

- Led the board of a student guild with 1,000 members, chaired board meetings, and managed administration

ARKAD Career fair*Company Host***Lund, Sweden**

Oct. 2022 – Nov. 2022

- Was responsible for two companies at the career fair (Gapwaves and Axis Communications)

SKILLS, PROJECTS & OTHER

Technical Skills: Python (PyTorch), Java, C++, C#, Scala, Haskell, SQL, CI/CD, Git, ROS2, Isaac Lab**Languages:** Swedish native, English C2 (IELTS Academic 8.5/9.0), Spanish A2, learning French**Personal Projects:** Created a VR simulation of a student dorm in Unreal, built an automated candy dispenser with a Raspberry Pi, developed an algorithm for solving Ricochet robots in Java, built an NLP model with Keras, developed regression models for biological experiments**Open-Source:** Working on an implementation of a more stable geodesic distance loss function in the PyRoMA framework for geometric deep learning with PyTorch**Media coverage:** [Dalslänningen](#), [Helsingborgs Dagblad](#)**Awards:** Won the national qualifiers and received a bronze medal at the European Union Science Olympiad (2019)